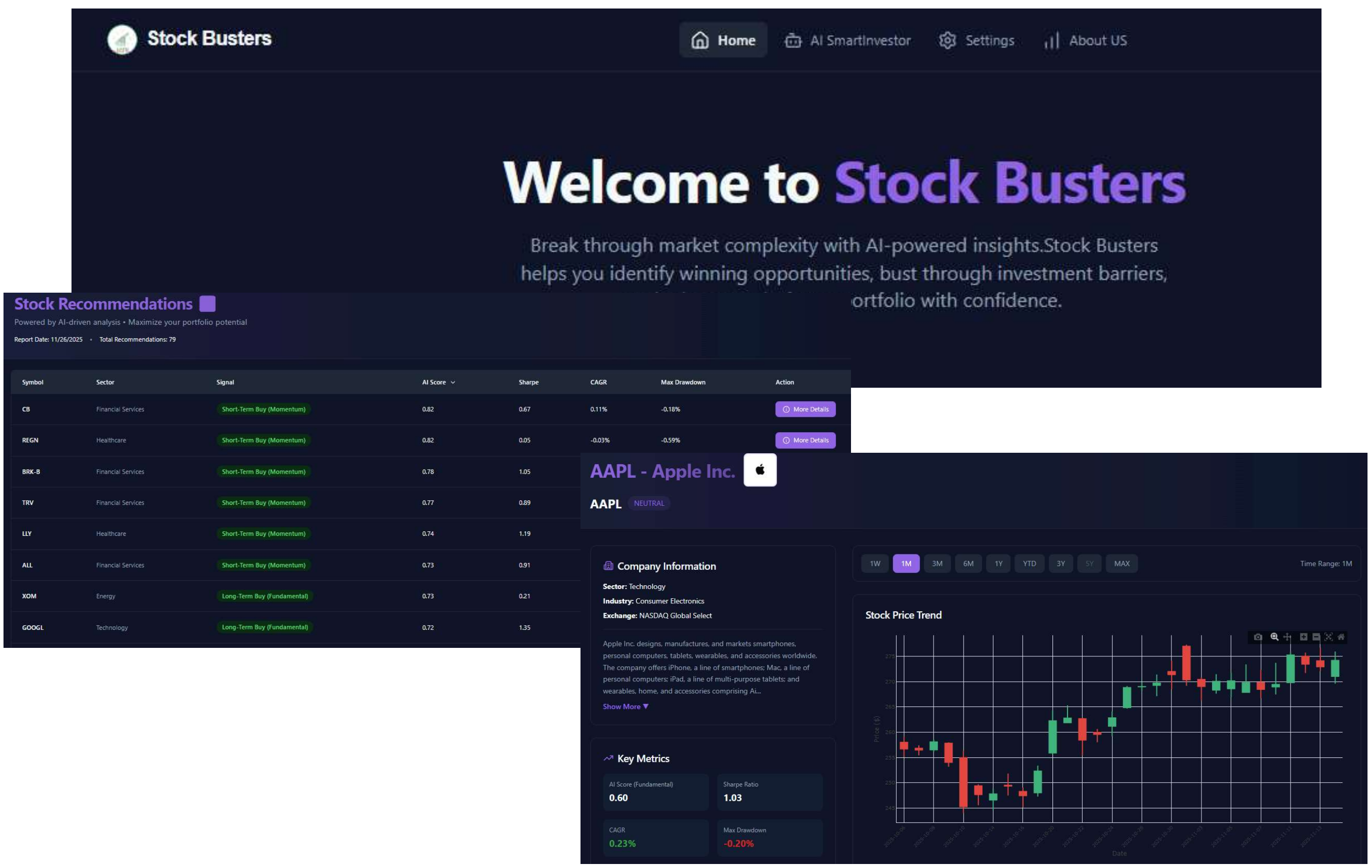
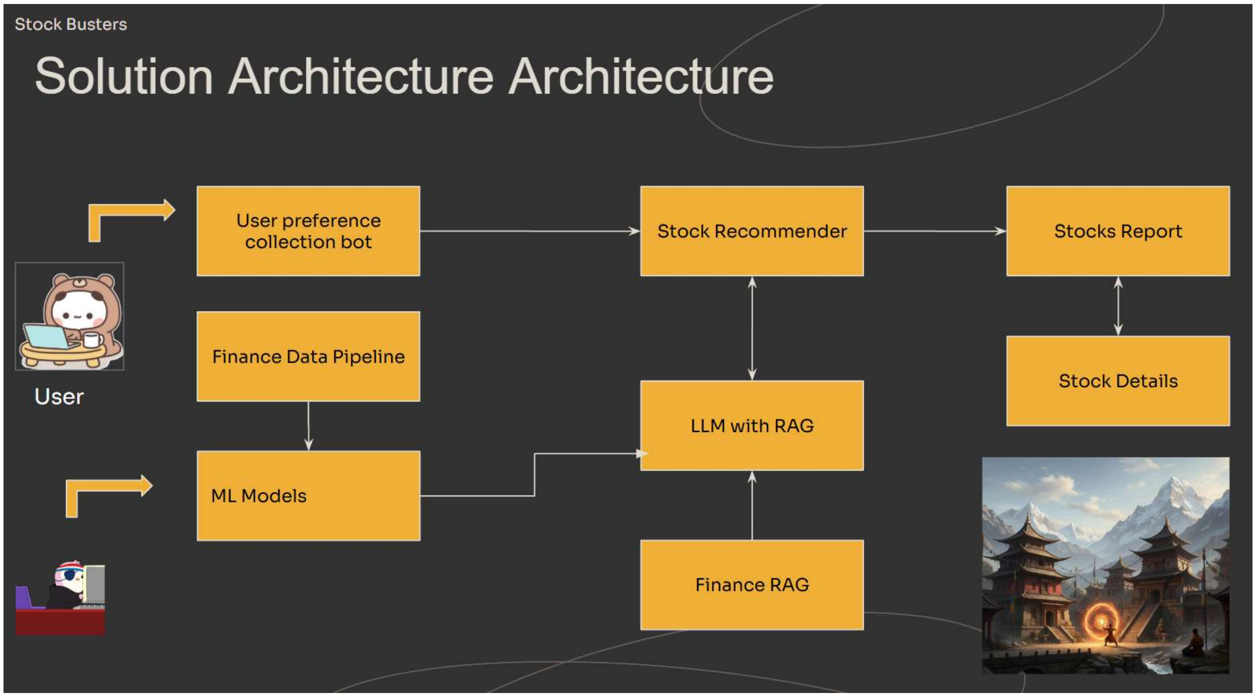
STOCKBUSTERS

ARCHITECTURE DESIGN DOCUMENT



SOLUTION ARCHITECTURE



User Preference Collection Bot

This module handles direct interaction with the User to gather personalized investment criteria, risk tolerance, investment horizon and other factors necessary for tailored recommendations. The bot is capable of answering clarifying questions while making the financial choices. This input stream feeds directly into the Stock Recommender.

Finance Data Pipeline

This is the core data ingestion engine. It is responsible for sourcing, cleaning, and processing raw financial data (e.g., historical prices, company information, sector information). The output of this pipeline feeds the ML model.

ML Models

These models consume the data from the Finance Data Pipeline. The model consumes both technical and fundamental data to make decisions. This fundamental includes metrics like ROE, ROIC, FCF Yield, Debt to Equity, cash, revenue, just to name a few. The technical data includes opening and closing prices, along with volume, high and lows for the day. With this information the model computes intermediate metrics which are used to come up with the final recommendation and ranking recommendations based upon fundamental and technical data.

LLM with RAG (Retrieval-Augmented Generation)

This is the system's reasoning and synthesis engine. It combines the model output and generates the reasoning for recommending a particular stock making use of expert knowledge from the RAG system. The LLM acts as an intermediary, providing enriched context and explanations to the Stock Recommender.

Recommendation and Reporting

These modules utilize the collected data and generated intelligence to produce the final user-facing results. The recommendations are correlated with the user preferences and tailored to their financial profile.

Stock Recommender

This is the central decision-making unit. It synthesizes three critical inputs; User Preferences, model output and LLM Context to create personalized recommendations.

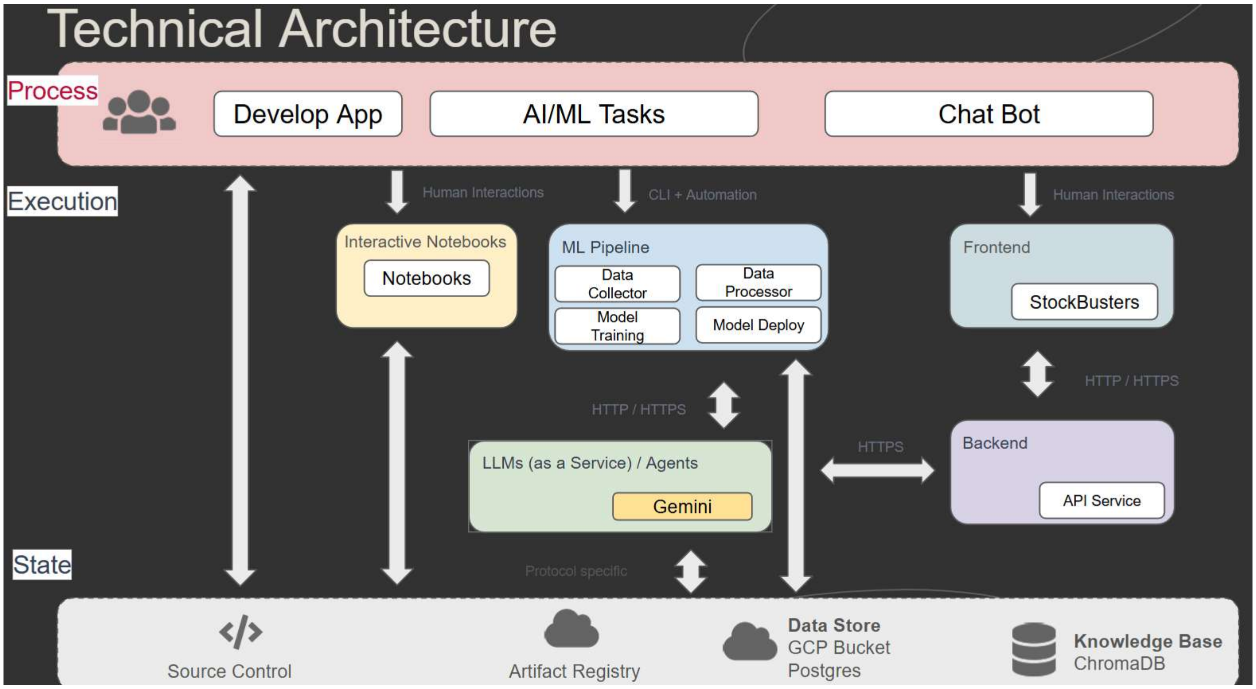
Stocks Report

The personalized recommendations are provided to the user in the form of stock recommendation report. This report summarizes the recommendations, including key performance indicators, risk assessments.

Stock Details

This module serves as a drill-down component. It provides granular information for any stock listed in the Stocks Report. This enlightens users with company information, the stock trend and rationale for recommending the stock.

TECHNICAL ARCHITECTURE



The architecture is designed to support application development, AI/ML tasks, and a Chat Bot feature, leveraging Large Language Models (LLMs) like Gemini.

Process Layer

This is the user and high-level function layer, representing the main areas of interaction and functionality supported by the system:

Develop App

Standard application development activities, interacting with the Execution and State layers.

AI/ML Tasks

Functions related to the core AI/ML capabilities, such as model development and training.

Chat Bot

The conversational interface for users,a key feature of the Stock Busters app, which involves Human Interactions to collect financial preferences for the user

Execution Layer

This layer contains the runtime components and services that handle the application's logic, processing, and user interaction:

Interactive Notebooks (Notebooks)

Used for human interaction, mainly data scientists or developers, for experimentation and development of AI/ML models. These connect to LLMs and the State layer.

ML Pipeline

An automated process for managing the entire machine learning lifecycle from collecting data to generating recommendations

Data Collector

Data collector collects the data through API from the datasources.

Data Processor

Prepares data for training or inference. This include preprocessing of the data and data cleanup
Model Training
The AI/ML model will be trained based on the data collected and processed in the data pipeline.

Model Deploy

The trained model is deployed into a production environment.

Agents Layer

The Agents work together in a graph network to collect the user preferences and convert the natural language in the right data objects to apply to the model

LLMs (as a Service)

Gemini is used here with the support of the RAG system. It uses financial knowledge to describe why the model has selected a particular stock.

Frontend (StockBusters)

The user-facing component of the main application, supporting "Human Interactions" and communicating with the Backend via HTTP/HTTPS.

Backend

The core application logic and data-handling services, accessible via HTTPS.

API Service

Handles business logic and serves the Frontend and LLMs.

Vector DB Service

Provides a vector database, holding financial knowledge base for the LLM

State Layer

This is the data and infrastructure layer that stores, manages, and tracks all persistent assets and data:

Source Control

Stores all application code, configuration, and potentially pipeline definitions.

Artifact Registry

Stores built artifacts, such as trained models from the ML Pipeline

Data Store

Stores raw, processed, and training data utilized by the ML Pipeline and LLMs.

Knowledge Base

Stores finance knowledge and in a Vector DB (Chroma DB), that is used by the LLM for reasoning